Test of the new Vialux DMD mount for holographic applications

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Abstract

In a previous tutorial, we discussed mechanical instabilities in holographic setups when using a Vialux DMD. The main issues where the lack of dedicated mount and the fact that the DMD board is connected to the PCB by a rigid flat cable, allowing 1/ applying a small torque on the DMD when handling the board, and 2/ transmitting vibrations originating from the PCB fan to the DMD. Recently, Vialux released a dedicated mount for their DMDs. In this report, we present a comparative analysis of different mounts strategies when using a DMD in a holographic setup, including the new Vialux mount and homemade solutions with and without foam damping.

Disclaimer

The mount was provided by Vialux for testing purposes, but this report is an independent analysis of the mount's performance in a holographic setup. Vialux did not influence the experimental design, data collection, analysis, or interpretation of results. Conclusion are our own, we did not receive any compensation for this work and are not influenced by any commercial interests.

1 Introduction

Holographic interferometric systems are intrinsically sensitive to optical perturbations but also mechanical vibrations. In lab experiments, we typically use devices and mounting hardware designed for minimizing vibrations. However, DMDs are not originally designed for holographic applications, but rather for display and projection purposes, where mechanical stability is not a critical requirement. In particular, the DMDs did not come with a dedicated mount, and the DMD board is connected to the PCB by a rigid flat cable, allowing for the transmission of vibrations from the PCB fan to the DMD. We showed in a previous tutorial that this can lead to significant phase fluctuations.

Recently, Vialux released a dedicated mount for their DMDs (see Fig. 1), which, according to Vialux, is *applications in a laboratory environment.* The goal here is to test the stability of the Vialux mount in a holographic setup and compare it to homemade solutions with and without foam damping, which are commonly used in our lab for mounting optical components.

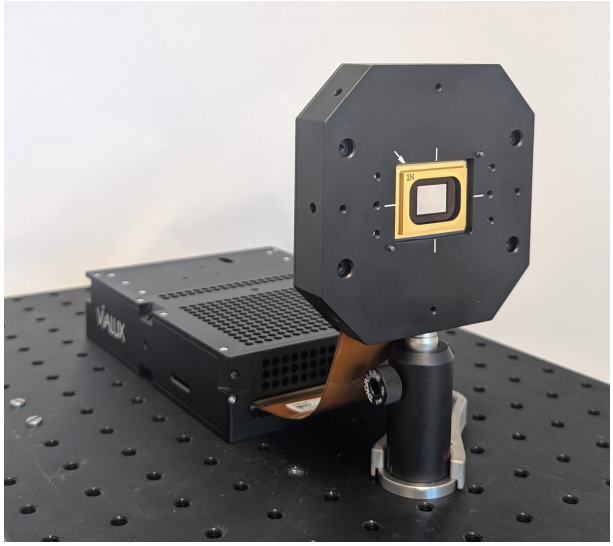

Figure 1: **Vialux DMD mount.** Images from the Vialux website.

1.1 Setup

We use as a source a 632 nm stabilized HeNe laser (Thorlabs HRS015B) coupled to a single mode fiber splitter. One beam is expanded, collimated and projected onto the DMD, while the other beam is used as the reference. The DMD diplays a static plane pattern, and the reflected beam is recombined with the reference beam to form the holographic interference pattern on the camera to form fringes. The setup is shown Fig. 2(left), and an example of the recorded interference pattern is shown in Fig. 2(right).

In all experiments, the optical layout remains strictly identical except for the element placed in the object arm. Depending on the experimental condition, this element is either a mirror or a digital micromirror device (DMD) mounted using different mechanical configurations.

The interference pattern is recorded using a CCD camera at 40 fps over approximately 180 s per acquisition. The recorded holograms are processed to extract the temporal phase fluctuations using the same approach as in this tutorial.

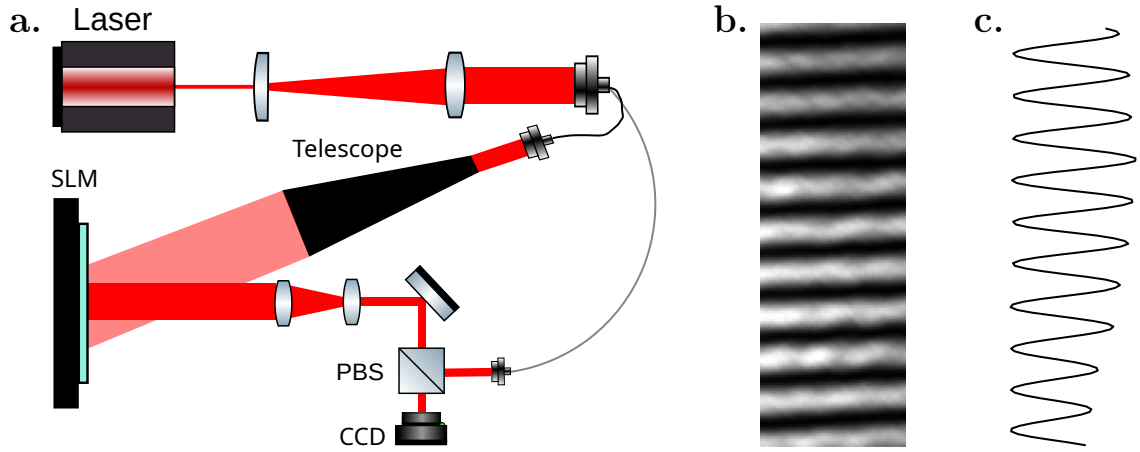

Figure 2: **Phase fluctuation measurement setup.** (a.) Schematic of the holographic experimental setup. The laser output is split into two beams using a fiber splitter. The object beam is reflected either by a mirror or modulated by a spatial light modulator (SLM, DMD), while the reference beam propagates unchanged. The two beams are recombined using a polarizing beam splitter (PBS) and interfere on a CCD camera, producing a holographic fringe pattern. (b.) Example of the recorded interference pattern (linear fringes) used for phase extraction. The stability of these fringes over time is analyzed to estimate phase fluctuations. (c.) 1D projection of the interference pattern along the direction of the fringes, used for phase extraction.

1.2 Experimental configurations

To establish a reference for the intrinsic stability of the optical setup, phase fluctuations are first measured using a mirror in the object arm. This configuration provides a baseline representing the best achievable stability under identical optical conditions. We first test our custom mouht, made of bulky steam scraps, with and without foam damping, as well as the Vialux mount with and without foam damping. Foam is used to clamp the flat cable to reduce vibrations (see here for more details). Then we test the Vialux mount. As it present lateral holes for mounting, we also test the effect of adding lateral stabilizers to the Vialux mount (see Fig. **??**), with and without foam damping. These configurations are summarized in Table 1, and configurations 2 and 5 are shown in Fig. 3.

ID	Configuration	Description
0	Mirror	Reference configuration (ground truth stability)
1	Custom mount	DMD mounted on a homemade mechanical holder
2	Custom mount + Foam	Homemade mount with foam damping to reduce vibrations
3	ViALUX mount	Commercial ViALUX LabKit DMD mount
4	ViALUX mount + Foam	ViALUX LabKit with additional foam damping
5	ViALUX mount + Foam + LS	ViALUX LabKit with foam damping and lateral stabilizers

Table 1: Summary of the experimental configurations investigated. Only the object-arm element and its mechanical mounting are modified between configurations.

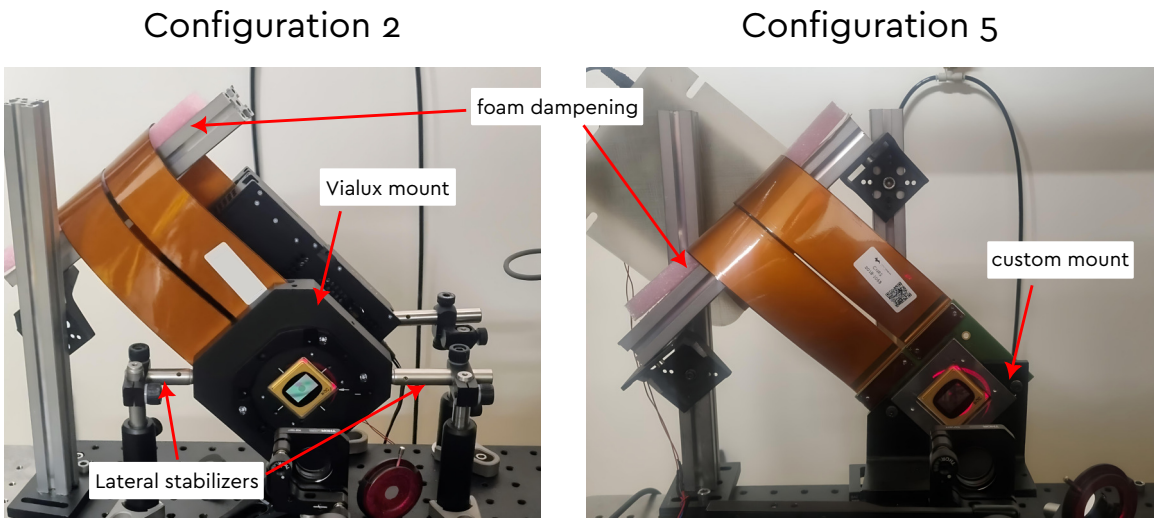

Figure 3: **Different mounting configurations.** (left) Configuration 2: Custom steal mount with foam damping. (right) Configuration 5: Vialux mount with foam damping and lateral stabilizers.

2 Methods

2.1 Phase Extraction

We measure interference patterns for 180 ss, 5 times for each configuration. In an ideal stable setup, the interference fringes should remain perfectly static overtime. We use the motion of the fringes to estimate the phase fluctuations, following the same approach as in this tutorial to extract the phase fluctuations from the recorded holograms. The main steps are as follows:

1. A region of interest (ROI) containing the holographic fringes is selected (see Fig. 2 right),
2. images are summed along the direction of the fringes to obtain 1D curves (see Fig.XXX),
3. the autocorrelation of the 1D curves for the first image is computed to find the period d of the fringes, corresponding to a phase shift a 2π,
4. we compute the intercorrelation between the first curve and each one at time t to find the position of the main peak, which corresponds to the distance $\Delta x(t)$ the fringes have moved compared to the initial position. This distance is directly related to the phase shift. The phase shift $\phi(t)$ is then computed as:

$$\phi(t) = 2\pi \frac{\Delta x(t)}{d} \tag{1}$$

2.2 Standard Deviation Analysis

For each configuration, and for each of the 5 runs, we compute the standard deviation of phase fluctuations $\phi(t)$.

2.3 Frequency Analysis

We compute the Fourier transform of the phase fluctuations $\phi(t)$ to analyze the frequency content of the vibrations affecting the system. We average the square amplitude of the Fourier transform across the 5 runs to obtain an average power spectral density (PSD) for each configuration.

3 Results

3.1 Phase Stability

Figure 4 shows the phase deviation over time for all conditions. The error bars indicate the range between minimum and maximum standard deviation among the five acquisitions.

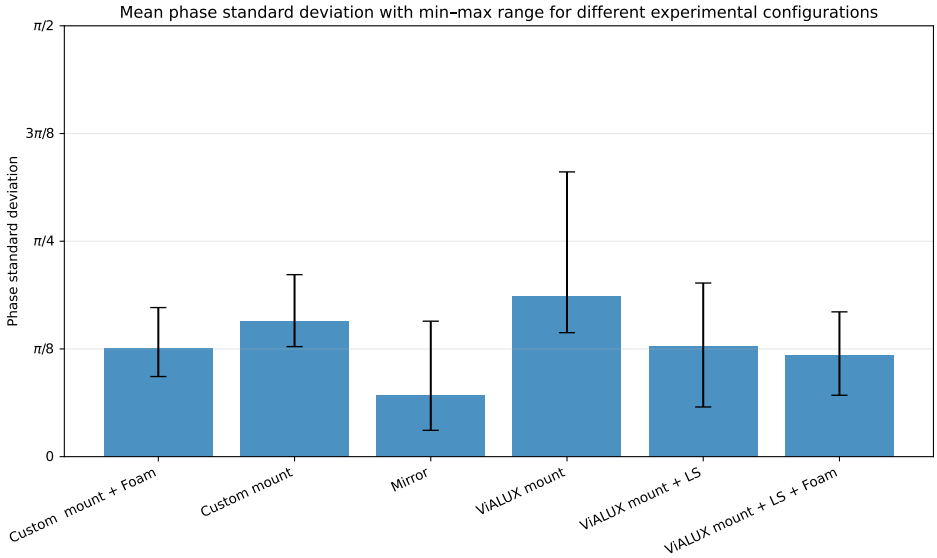

Figure 4: Mean phase deviation with min/max error bars for all experimental conditions.

We observe significant variations over the 5 runs, which means that we have other sources of fluctuations, such as air turbulence. Within the error bars, we see that the performance of the Vialux mount is in par with the homemade mount, and that adding lateral stabilizers to the Vialux mount reduces even more phase fluctuations.

3.2 Frequency Domain Analysis

As the standard deviation analysis is not very precise due to the large variations between runs, we also analyze the frequency content of the phase fluctuations by computing average PSDs for each configurations. Results are summarized in Fig. **??**.

Note that it is likely that the 40Hz acquisition rate is not sufficient to capture the full spectrum of mechanical vibrations, the goal is to qualitatively compare the dominants frequency components between the different configurations, to assess fluctuation reduction.

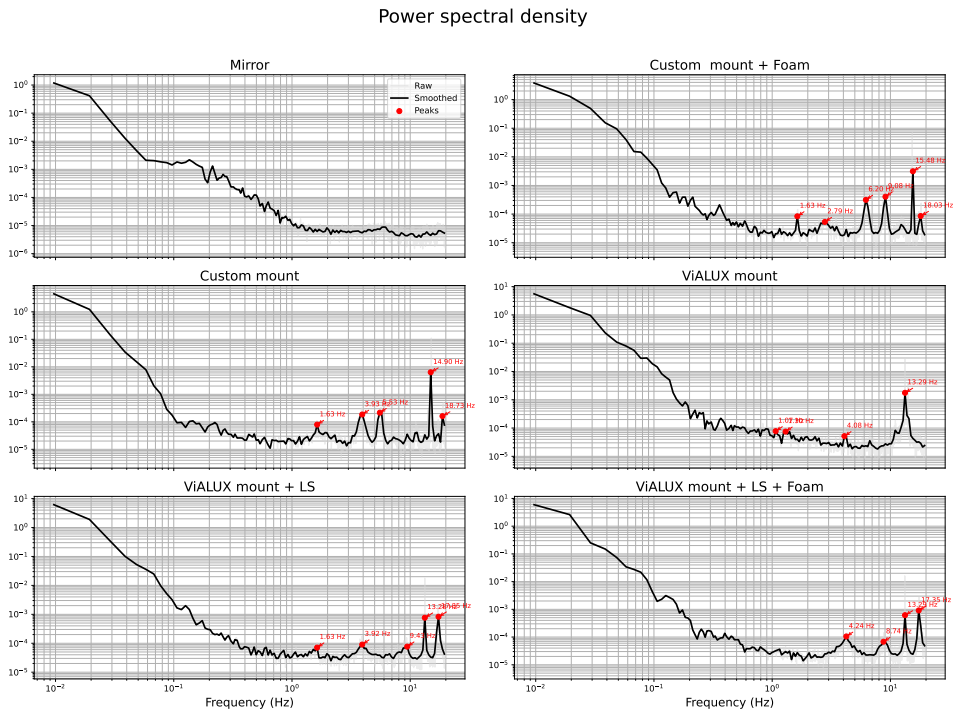

Figure 5: **Power spectral density (PSD).** Average PSDs for all experimental conditions, with characteristic peaks labeled.

belfig:psd

We now observe a clear reduction of high frequency components of the phase fluctuations when using the Vialux mount, and even more when adding lateral stabilizers.

4 Conclusion

While a more quantitative analysis with a faster acquisition rate would be needed to fully characterize the performance of the Vialux mount, our qualtative analysis suggest that the Vialux mount provides better stability than homemade solutions. While the difference is not huge, we did spend a lot of time and many iteration to optimize our homemade mount, and the Vialux mount provides a ready-to-use solution that is likely to be more stable and less sensitive to handling.